

Auto-Decte: An LLM- and RAG-Assisted Human-in-the-Loop Document Intelligence System for Industrial Paper Forms

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Abstract—Deploying large language models (LLMs) in high-stakes enterprise workflows is constrained less by model capability than by trust: suggestions must never silently become facts, every decision must be traceable to evidence, and the system must keep working when the model is unavailable. This paper presents Auto-Decte, a document intelligence system for industrial paper forms that operationalizes these constraints through a human-in-the-loop architecture. A fiducial-guided perception front end (QR/ArUco template recognition, perspective correction, per-cell digit and OMR recognition with an explicit ambiguity band) converts photographed paper forms into structured candidate values. An LLM suggestion adapter (DeepSeek, structured output) and a local vector similarity-retrieval module (RAG-style, read-only) assist reviewers, while a React review workbench makes the human the only source of truth: machine outputs are stored as append-only candidates, and only human confirmations or corrections become versioned, auditable records. A PWA front end with an offline outbox extends submission to in-plant mobile devices. The system is evaluated on end-to-end acceptance (389 public-snapshot tests, including the AI-off path and suggestion-isolation invariants), front-end recognition benchmarks (digit 93.9% overall, 100% on the printed font family; OMR 100% correct outside the ambiguity band; QR 100%; ArUco correction ≤ 0.34 px), a baseline comparison against a learned HOG+SVM recognizer (comparable accuracy, but only the template matcher routes ambiguity to humans), end-to-end whole-form cell accuracy of 97.5% clean, and per-image latency (≈ 2.2 s, QR search dominant).

Index Terms—LLM application, retrieval-augmented generation, human-in-the-loop, document intelligence, traceability, offline-first PWA

I. INTRODUCTION

Foundation-model-assisted document processing is attractive for the long tail of paper-based workflows in small and medium plants, but its adoption in payroll-adjacent settings is blocked by a trust problem: an LLM that transcribes a digit wrong changes a worker’s wage, and a retrieval system that injects a similar-looking historical record into a decision can silently corrupt it. Fully automatic pipelines are therefore unacceptable where errors carry financial consequences, yet purely manual digitization does not scale.

Auto-Decte addresses this with a human-in-the-loop architecture built on three principles: (i) *machine outputs are candidates, never facts* — the LLM adapter and every recognition

stage emit candidates that a human must confirm or correct; (ii) *the document is engineered* — templates are generated with QR codes and ArUco fiducials so that perception is reliable rather than learned; (iii) *everything is auditable* — evidence is content-addressed, records are append-only versions, and every exported cell is back-traceable. The system is deployed as a working baseline at a bamboo-processing plant.

Contributions:

- 1) A human-in-the-loop document intelligence architecture for industrial paper forms, integrating a fiducial perception front end, an LLM suggestion adapter, and local similarity retrieval under a single “candidates-only, human-confirmed” trust boundary.
- 2) Trustworthy LLM/RAG integration mechanisms: suggestions are stored separately and can never mutate confirmed values; retrieval results are strictly read-only; the core pipeline runs and remains correct with the AI layer fully disabled (guarded by acceptance tests).
- 3) An engineered, interpretable perception front end — synthetic-font template matching for digit cells and a fill-ratio OMR recognizer — whose explicit ambiguity band routes borderline cells to humans instead of guessing.
- 4) An offline-capable mobile submission path (PWA + IndexedDB outbox + idempotency keys) that feeds the same review queue as scanned forms.
- 5) A deployment-motivated case study with end-to-end acceptance evidence and honest reporting of what remains unmeasured.

II. RELATED WORK

Document OCR and layout analysis. General document AI stacks (PP-OCRv3 [1], LayoutLMv3 [2], Donut [3]) target free-form documents with learned detection and recognition. Auto-Decte deliberately keeps learned layout models out of the critical path: system-generated templates make layout known exactly, so perception reduces to fiducial-guided cropping plus per-cell template matching; learned OCR remains a future extension for handwriting and Chinese text.

LLM assistance and retrieval. Retrieval-augmented generation and generative information-extraction surveys [9], [10]

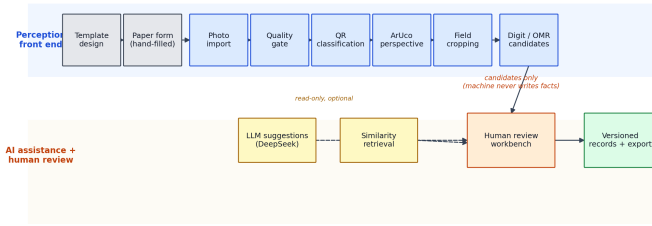


Fig. 1. End-to-end pipeline. Recognition and the LLM produce candidates and references; only human confirmations or corrections become facts.

and Local-First Software principles [11] motivate two components: a read-only DeepSeek structured-output suggestion adapter and a local vector similarity index. The distinction from generic RAG is architectural: in Auto-Decte the model and the retriever are denied write access to facts by construction, and their outputs are versioned separately from records.

Human-in-the-loop digitization. HITL principles for high-stakes data (Budd et al. [7]; drilling-report digitization [8]) motivate the review workbench: machine outputs are append-only candidates, and every confirmed or corrected value is a versioned human decision with a before/after audit pair.

Fiducial-based localization and OMR. ArUco [5] and MILP dictionary markers [6] provide geometric anchoring; existing OMR systems (e.g., SurveyNet [4]) evaluate bulk mark recognition. Auto-Decte’s distinguishing feature is the first-class ambiguity band: cells in the intermediate fill band are emitted with zero confidence and force review rather than a thresholded guess.

Positioning. Four decisions distinguish Auto-Decte from prior systems: (i) machine outputs — from recognition and from the LLM alike — are candidates, never facts; (ii) an explicit ambiguity band routes borderline cells to humans; (iii) full traceability — SHA-256 evidence, versioned records, audit events, back-traceable exports; (iv) offline mobile submission integrated with the same review queue.

III. SYSTEM OVERVIEW

Auto-Decte is a single-machine deployment: FastAPI (Python 3.11) backend, SQLite via SQLAlchemy 2.0 with Alembic migrations, React 18 + Vite front end, OpenCV for image operations, a DeepSeek client for structured-output suggestions, and a local vector index for similarity retrieval. Fig. 1 shows the pipeline stages.

The domain model separates immutable evidence, versioned form records (`RecordVersion`), append-only recognition attempts, isolated AI suggestions, read-only retrieval references, audit events, and export batches. Roles (operator, reviewer, finance, admin, auditor) gate every mutation.

IV. TEMPLATE DESIGN AND PERCEPTION FRONT END

Templates are designed and published inside the system (template center with versioning and draft canvas). Each published template renders a PDF/PNG with a QR code (template identity + version) and ArUco markers at known positions.

Recognition is fiducial-first: a multi-region QR search classifies the form, four ArUco corner markers (IDs 10–13) map it to the canonical canvas, perspective correction aligns it, and per-field cells are cropped from the template definition. This makes identification and geometry deterministic and auditable.

V. FIELD-LEVEL RECOGNITION WITH CONFIDENCE AND AMBIGUITY

Digit cells. Each cell is normalized to 48×64 , binarized with Otsu thresholding, and matched against ten rendered synthetic-font templates (0–9) by normalized mean absolute difference. The recognizer reports the best digit, a distance-based confidence, and an `AMBIGUOUS` flag when the margin to the second-best template falls below 0.02. Blank cells (ink ratio $< 1\%$) are reported as blank rather than as a digit.

Checkbox (OMR) cells. Fill ratio in the cell interior is computed after Otsu binarization; ratios ≤ 0.15 map to unchecked, ≥ 0.35 to checked, and the $(0.15, 0.35)$ band maps to `AMBIGUOUS` with zero confidence. The ambiguity band routes exactly the cells that recognition cannot decide to the human, instead of silently guessing.

VI. LLM ASSISTANCE AND SIMILARITY RETRIEVAL

Two optional AI components support the reviewer, both constrained to be unable to write facts: (i) *LLM suggestions* — a DeepSeek adapter with structured output requests field suggestions; suggestions are stored separately from records, carry their evidence reference, and can never mutate confirmed values, and the adapter is disabled by default so the core pipeline runs without any external model (an acceptance-tested AI-off path); (ii) *similarity retrieval* — a local vector index shows similar historical forms as reference, and retrieval results are strictly read-only (guarded by acceptance tests). The trust boundary is architectural: the model and the retriever produce candidates and references; only the reviewer produces facts.

VII. DATA INTEGRITY, VERSIONING, AND EXPORT

Evidence images are content-addressed with SHA-256; duplicates are detected and never silently overwritten. Every human correction appends a `RecordVersion` with before/after audit rows. XLSX export produces four worksheets (formal data, exceptions & review, summary, export notes) and every exported cell is back-traceable to form, version, evidence image, and event log.

VIII. OFFLINE MOBILE SUBMISSION

A PWA front end supports in-plant phone use: mobile credentials are salted with scrypt; sessions use `HttpOnly SameSite` cookies with CSRF double-submit; drafts live in an IndexedDB outbox keyed by owner/device with idempotency keys; retries back off on network errors and stop on 4xx conflicts; mobile submissions flow into the same `NEEDS_REVIEW` path as scanned forms.

IX. EVALUATION

Methodology: end-to-end functional acceptance via the automated test suite; controlled benchmarks on the perception front end using the system’s canonical template conventions and synthetic perturbations (blur, rotation, brightness, Gaussian noise, perspective tilt) plus fill-ratio sweeps for OMR; per-stage latency on A4@150 dpi images (desktop hardware, CPU timing).

Functional acceptance and trustworthy-AI invariants.

On the public snapshot of the integration branch, 389 tests pass. Beyond core digitization (import, SHA-256 dedup, QR classification, digit/OMR candidates, rule validation, human confirmation, export, reverse traceability), the suite enforces the AI trust boundaries: the AI-off path completes the workflow end-to-end; LLM suggestions are stored separately from confirmed values; similarity-retrieval results are read-only references. 159 additional tests require enterprise database fixtures excluded from the public repository by data policy.

Perception front end. (i) Digit recognition: 93.9% overall across 28 font×perturbation conditions; 100% on the system’s printed font family under blur/rotation/noise/brightness; 80–90% on unseen fonts with 10–20% of cells flagged ambiguous. (ii) OMR: ratios ≤ 0.15 unchecked (100% correct), ≥ 0.50 checked (100% correct), (0.15, 0.50) ambiguous (100% routed to review). (iii) Template detection & geometry: QR recovery 100% on 90 perturbed renders; ArUco correction 100% under blur/noise/brightness/tilt with ≤ 0.34 px alignment error, dropping to 50%/20% at 3°/5° rotation. (iv) Latency: QR search 2,180 ms, quality assessment 20 ms, perspective correction 35 ms per image.

Baseline comparison. A learned HOG + linear-SVM recognizer trained on the printed font is compared against the template matcher under the same 28 conditions. On the printed font family both reach 100%; on unseen fonts the two are comparable (SVM 79.7–80.3% vs template 75.4–79.7% confirmed, with the template matcher additionally flagging 10–15% of cells as ambiguous). The differentiator is not raw accuracy but the failure mode: the SVM has no ambiguity mechanism, so its errors are silent, whereas the template matcher routes borderline cells to the reviewer.

End-to-end whole-form evaluation. Synthetic filled forms (10 digit cells with random printed fonts + 10 OMR boxes) pass the full pipeline (QR classification → ArUco perspective correction → field cropping → per-cell recognition). Cell-level accuracy is 97.5% clean, 96.5–98.0% under blur/noise/brightness/tilt, with 4.5–8.5% of digit cells routed to human review; OMR accuracy is 94–100%. Under 3° rotation the form-correction success drops to 60% and accuracy to 57.5%/60.0%, consistent with the fiducial recovery’s known rotation sensitivity.

LLM/RAG effectiveness (functional, not yet quantitative). The current release establishes correctness and isolation of the AI layer, not yet its efficiency benefit; a quantitative reviewer-time-reduction study on a 100–300-form field pilot is future work.

X. LIMITATIONS AND FUTURE WORK

Honest scope limits: general handwritten digit-string / Chinese text / signature OCR is not implemented (only single-cell template digits and OMR); deployment is single-machine without a completed multi-user capacity test or field pilot; the evaluation lacks a 30–50-form golden set with double entry and real plant images, so accuracy claims are limited to controlled benchmarks; and the LLM suggestion module has functional correctness but no measured end-to-end efficiency gain. Future work: general OCR integration, independent task workers, external storage, a quantitative LLM-assistance study (reviewer time, error-catch rate, suggestion acceptance rate), and a field pilot measuring financial-processing time reduction.

XI. CONCLUSION

Auto-Decte demonstrates a trust-first integration of an LLM adapter and similarity retrieval into an industrial document workflow: recognition and the model produce candidates and references, while humans remain the only source of facts, with full traceability from photograph to exported cell. The system is deployed as a working baseline at a bamboo-processing plant and provides the substrate for a future quantitative field study.

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